# Topic 3 — Nutrition, Vitamins, Diseases and Medicine

**Covers syllabus (click to expand)**

Human Body Composition & Major Elements | Nutrition & Balanced Diet (Macronutrients, Micronutrients, Energy Values) | Carbohydrates (Sugars: Glucose, Fructose, Sucrose, Starch, Glycogen, Cellulose) | Proteins, Essential Amino Acids & Protein-Energy Malnutrition (Kwashiorkor vs Marasmus) | Enzymes as Biocatalysts | Fat-Soluble Vitamins (A, D, E, K) & Water-Soluble Vitamins (B-Complex, C) | Vitamin Chemical Names, Rich Dietary Sources & Deficiency Syndromes | Macro-minerals, Micro-minerals & Trace Elements | Toxic Heavy Metals & Environmental Poisoning (Minamata, Itai-Itai, Blue Baby Syndrome, Arsenic, Fluorosis) | Classification of Human Diseases (Infectious vs Non-Infectious, Communicable vs Non-Communicable) | Bacterial Diseases & Tests (TB, Typhoid, Cholera, Leprosy, Diphtheria, Tetanus, Plague) | Viral Diseases (Smallpox, Polio, Rabies, Dengue, Chikungunya, Hepatitis, HIV/AIDS, Nipah, COVID-19) | Protozoan & Helminthic Diseases (Malaria, Kala-azar, Sleeping Sickness, Filariasis) | Cancer Biology (Carcinoma, Sarcoma, Leukemia, Lymphoma, Diagnostic Biomarkers) | Diabetes Mellitus (Type 1 & 2, Insulin, HbA1c) vs Diabetes Insipidus (ADH) | Vaccines & Immunisation (Live Attenuated, Inactivated, Toxoids, mRNA, Viral Vectors) | Diagnostic Tests & Clinical Instruments

**Sources baked in:** NCERT Class 11 & 12 (Biomolecules, Human Health & Disease), WHO disease fact sheets, ICMR Dietary Guidelines, UPPCS Prelims PYQs (2018–2025) and UKPCS Prelims **Weight:** ★★★ High — heavily tested: vitamin deficiencies, match-pairs of diseases and causative pathogens, toxic metal poisonings, COVID vaccine technologies, diabetes mechanisms, and diagnostic tests **Last verified:** September 2026

## Current Affairs

| Year | Event / Fact | Must-Score Fact |
|---|---|---|
| **2024–25** | WHO documented recurring **Nipah virus** cases in Kerala, localized to **Malappuram and Palakkad** districts. | Nipah virus is a zoonotic Henipavirus transmitted by fruit bats (genus ***Pteropus***); causes fatal acute encephalitis; **not** mosquito-borne. |
| **2025** | Global malaria eradication push: WHO prequalified the second malaria vaccine **R21/Matrix-M** (developed by University of Oxford & Serum Institute of India). | Follows RTS,S/AS01 (Mosquirix); both target the circumsporozoite protein of *Plasmodium falciparum*. |
| **Still Current** | **Sickle Cell Anaemia Elimination Mission** launched by Prime Minister from Shahdol, Madhya Pradesh, targeting elimination by **2047**. | Focuses on tribal populations; point mutation on chromosome 11 substituting Glutamic acid with Valine ($GAG \rightarrow GUG$). |

| Year | Event / Fact | Must-Score Fact |
|---|---|---|
| **COVID Platform Facts** | Diverse COVID-19 vaccine platforms: **Covaxin** = Inactivated whole virus; **Covishield** = Non-replicating chimpanzee adenovirus vector; **Moderna / Pfizer** = mRNA nanoparticle; **Corbevax** = Protein subunit; **ZyCoV-D** = Plasmid DNA (world's first needle-free DNA vaccine, Zydus Cadila). | Frequently tested matching pair in state PSCs (UPPCS 2022 tested Sinovac = inactivated whole virus). |

## 3.1 Human Body Composition & Macronutrient Dynamics

- **Chemical Elemental Composition of the Human Body**:
- The human body is composed of 6 primary major bio-elements accounting for **~99% of total body mass**:
  - **Oxygen ($\sim 65\%$)**: Most abundant element by mass due to the vast water content ($\sim 60$–$65\%$ of body weight).
  - **Carbon ($\sim 18.5\%$)**: Backbone of all organic macromolecules (carbohydrates, lipids, proteins, nucleic acids).
  - **Hydrogen ($\sim 9.5\%$)**: Most abundant element by **total atom count** (each water molecule $H_2O$ contains two hydrogen atoms).
  - **Nitrogen ($\sim 3.2\%$)**: Integral constituent of proteins, amino acids, DNA, RNA, ATP, and creatine.
  - **Calcium ($\sim 1.5\%$)**: Most abundant mineral; structural hydroxyapatite in bone and teeth, blood coagulation, muscle contraction.
  - **Phosphorus ($\sim 1.0\%$)**: Calcium phosphate in bones, phospholipid membranes, phosphodiester bonds of DNA/RNA, and high-energy ATP.
- **Macronutrients: Carbohydrates, Proteins & Fats**:
- **Carbohydrates**: Primary and immediate source of physiological energy; stored as **Glycogen** in liver (~100 g) and skeletal muscles (~400 g). Excess carbohydrates are converted to neutral fats via **lipogenesis** and deposited in subcutaneous adipose tissue.
- **Proteins**: Primary structural "building blocks" of body tissues; consist of 20 standard L-$\alpha$-amino acids; required for growth, tissue repair, enzymes, peptide hormones, antibodies, and blood clotting factors.
- **Fats (Lipids)**: Concentrated energy reserves; structural components of cell membranes (phospholipids); carriers for fat-soluble vitamins (A, D, E, K); provide thermal subcutaneous insulation and mechanical shock absorption for internal organs.
- **Protein-Energy Malnutrition (PEM) — Kwashiorkor vs Marasmus**:
- Prevalent in developing countries among infants and young weaning children:
- **Kwashiorkor (Ghanian: "disease of deposed child")**:
  - Cause: **Severe deficiency of protein in the presence of adequate or fair carbohydrate calories** (child is switched from protein-rich breast milk to low-protein starch gruel).

    - Symptoms: Severe pitting **oedema** (swelling of limbs, feet, and face due to hypoalbuminaemia and oncotic pressure collapse), swollen protruding "pot belly", enlarged fatty liver, lethargy, muscle wasting masked by fluid retention, peeling **"flaky-paint dermatosis"**, and sparse reddish-orange hair.

- **Marasmus (Greek: "withering / wasting")**:
    - Cause: **Combined severe deficiency of both proteins AND total caloric energy**; commonly seen in infants under **1 year of age** when mother's milk is replaced too early by watery, nutritionally deficient foods.
    - Symptoms: Extreme emaciation, complete disappearance of subcutaneous fat; ribs become prominent; skin becomes loose, thin, and dry, giving the infant the shrivelled appearance of a **"little old man"** or "monkey face"; **oedema is strictly absent**.

## 3.2 Dietary Sources, Nutrient Chemistry & Food Adulteration

Dietary physiology evaluates the biological availability of nutrients, dietary standards, and toxic adulterants commonly tested in public service examinations:

### High-Yield Dietary Sources

- **Protein Concentrations in Common Foods:**
- **Soybean (*Glycine max*):** Highest protein concentration among plant foods (**~40–42% protein by weight**), containing all essential amino acids with high biological value.
- **Groundnut / Peanut:** Rich dual source of both **protein (~25%)** and **lipids/fats (~45%)**.
- **Pulses and Legumes (Gram, Pigeon Pea / Arhar, Lentil):** Average ~20–25% protein.
- **Wheat Dough Elasticity & Gluten:**
- The substance responsible for the bread-making quality and viscoelastic dough strength in wheat (*Triticum*) flour is **Gluten** (a composite storage protein composed of gliadin and glutenin). Individuals with Celiac disease suffer autoimmune intestinal damage upon consuming gluten.
- **Caloric Fuel Yields:**
- Carbohydrates: $\sim 4.0\ \text{kcal/g}$
- Proteins: $\sim 4.0\ \text{kcal/g}$
- Fats / Lipids: $\sim 9.0\ \text{kcal/g}$
- **Pure Water:** Yields **Zero (0) calories** (contains no macronutrient fuel).
- **Specialized Micronutrient Food Sources:**
- **Iron in Fruits: Karonda (*Carissa carandas*)** is the richest known fruit source of dietary iron (~39 mg / 100 g dry fruit), followed by date palm and green leafy vegetables.
- **Vitamin C (Ascorbic Acid):** Highest in **Aonla (Indian Gooseberry)**, followed by guava, citrus fruits (lemon, orange), and **green chilli** (*Capsicum frutescens*).
- **Vitamin D:** Endogenously synthesized in human epidermis upon exposure to **solar UVB radiation** from 7-dehydrocholesterol; dietary sources include cod-liver oil and fortified dairy.
- **Thyroxine is a Hormone:** *Exam Trap Alert*: Thyroxine is an iodine-containing tyrosine hormone secreted by the thyroid gland; it is **NOT a vitamin**.

- **Fenugreek Seeds (*Trigonella foenum-graecum* / Methi):** Clinically proven dietary supplement that normalizes blood glucose in diabetes mellitus due to high soluble fiber (galactomannan) and 4-hydroxyisoleucine stimulating insulin secretion.
- **Dairy & Cheese:** Fermented dairy products (cheese, paneer) are concentrated sources of casein protein, calcium, and fat; cheese also provides Vitamin A and Vitamin B12.
- **Starch Chemistry in Rice (Amylose vs Amylopectin):**
- **Amylose:** Linear $\alpha$-(1,4)-glucan polymer. Rice varieties with **high amylose content** cook into separate, dry, non-sticky grains.
- **Amylopectin:** Branched $\alpha$-(1,6)-glucan polymer. Rice varieties with low amylose and high amylopectin cook into sticky, glutinous clumps.

### Toxic Food Adulteration & Dietary Syndromes

- **Argemone Oil Adulteration & Epidemic Dropsy:**
- Mustard seeds and mustard oil are frequently adulterated with seeds of **Argemone mexicana (prickly poppy)**.
- Contains toxic alkaloids (**sanguinarine** and dihydrosanguinarine) that inhibit sodium-potassium ATPase, causing widespread vascular leakage, severe bilateral pitting edema, cardiac failure, and glaucoma, known clinically as **Epidemic Dropsy**.
- **Khesari Dal & Neurolathyrism:**
- Regular, prolonged consumption of **Khesari dal (*Lathyrus sativus* / grass pea)** induces **Neurolathyrism**—an irreversible, crippling neurodegenerative spastic paralysis of the lower limbs.
- Caused by the neurotoxin **$\beta$-ODAP** ($\beta$-N-oxalyl-$\alpha$, $\beta$-diaminopropionic acid), a structural analogue of glutamate that overstimulates and kills motor neurons in the spinal cord.
- **Botulism (Food-Borne Intoxication):**
- A lethal, acute neuroparalytic disease caused by consuming food contaminated with pre-formed **Botulinum neurotoxin** produced by the anaerobic spore-forming bacterium ***Clostridium botulinum***.
- Classified strictly as a **food-borne intoxication** (not an active bacterial infection), typically contracted from improperly sterilized, home-canned vegetables, meat, and vacuum-packed foods.

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## 3.3 Pathogen Classification, Clinical Syndromes & Genetic Disorders

### Organ Specificity of Common Bacterial & Viral Diseases

- **Tuberculosis vs Typhoid Organ Targets:**
- **Tuberculosis (*Mycobacterium tuberculosis*):** Primarily attacks the **Lungs (pulmonary TB)**, but can disseminate to lymph nodes, spine (Pott's disease), and meninges.
- **Typhoid Fever (*Salmonella typhi*):** Primarily attacks and ulcerates Peyer's patches in the **Small Intestine**, producing intestinal bleeding, perforation, and step-ladder remittent fever (diagnosed by Widal agglutination test).
- **SARS-CoV-2 (COVID-19):**
- The global pandemic of 2020–2022 was caused by **Severe Acute Respiratory Syndrome Coronavirus 2 (SARS-CoV-2)**, an enveloped, positive-sense single-stranded RNA virus entering cells

via the ACE2 (Angiotensin-Converting Enzyme 2) receptor.

- **Dengue Fever & Blood Platelets:**
- Caused by the Dengue flavivirus and transmitted by diurnal ***Aedes aegypti*** mosquitoes. Hallmark laboratory finding is sudden, severe **Thrombocytopenia (precipitous collapse of blood platelet count)**, predisposing to hemorrhagic fever and dengue shock syndrome.
- **Malaria Organ Target:**
- *Plasmodium* parasites invade and rupture red blood cells, causing severe hemolytic anemia and massive enlargement and congestion of the **Spleen (Splenomegaly)**.
- **Prion Diseases (Transmissible Spongiform Encephalopathies):**
- **Prions:** Infectious, misfolded protein molecules devoid of any nucleic acid (DNA or RNA).
- Cause fatal neurodegenerative diseases: **Bovine Spongiform Encephalopathy (BSE / "Mad Cow Disease")** in cattle, Scrapie in sheep, and Creutzfeldt-Jakob Disease (CJD) and Kuru in humans.
- **Fungal Dermatophyte Infections:**
- **Baldness (Alopecia / *Tinea capitis*):** Fungal infection of scalp hair follicles caused by *Microsporum* or *Trichophyton*.
- **Athlete's Foot (*Tinea pedis*):** Pruritic scaling fungal infection of web spaces of the feet caused by *Trichophyton mentagrophytes*.
- **Ringworm (*Tinea corporis*):** Circular, erythematous skin lesions with raised margins caused by dermatophytic fungi.
- **Anthrax (Woolsorter's Disease):**
- Acute zoonotic disease caused by the spore-forming rod ***Bacillus anthracis***.
- Primarily infects herbivorous animals (**sheep, goats, cattle**); humans acquire pulmonary anthrax by inhaling spores from contaminated animal hair and hides in wool-sorting factories.

### Occupational & Environmental Lung Diseases

- **Black Lung Disease (Coal Workers' Pneumoconiosis - CWP):**
- Occupational lung disease caused by long-term inhalation of respirable coal mine dust in the **coal mining industry**, leading to pulmonary fibrosis and progressive massive fibrosis.
- **Silicosis:** Caused by inhaling crystalline silica dust in stone quarries, sandblasting, and glass industries.
- **Asbestosis:** Caused by asbestos fibers in shipbreaking, mining, and insulation manufacture (predisposes to malignant mesothelioma).
- **Byssinosis ("Brown Lung"):** Caused by cotton dust inhalation in textile spinning mills.

### Genetic & Hereditary Disorders

- **Haemophilia ("Royal Disease"):**
- **X-linked recessive bleeding disorder** caused by deficiency of functional clotting factors: **Factor VIII** (Haemophilia A, ~80% of cases) or **Factor IX** (Haemophilia B / Christmas Disease).
- Transmitted through asymptomatic carrier females ($X^HX^h$) to their sons ($X^hY$). A carrier mother exhibits criss-cross inheritance, passing the defective gene to the **sons of her daughters**.
- **Down Syndrome (Trisomy 21):**

- Genetic chromosomal disorder caused by nondisjunction resulting in an extra copy of chromosome 21 (**47, XX,+21 or 47, XY,+21**). Manifests with mental retardation, flat facial profile, simian crease across palms, and cardiac septal defects.
- **Duchenne Muscular Dystrophy (DMD):**
- Severe X-linked recessive genetic disorder caused by mutations in the dystrophin gene on the X chromosome, leading to progressive degeneration of skeletal muscle fibers and premature death.
- **Colour Blindness:**
- X-linked recessive disorder causing failure of red-green retinal cone photoreceptors; affects predominantly males ($X^cY$).

### Metabolic & Behavioral Pathologies

- **Gout (Metabolic Arthritis):**
- Characterized by hyperuricemia (elevated serum **uric acid** levels $> 7.0\ \mathrm{mg/dL}$). Uric acid crystals (**monosodium urate**) deposit in synovial joints and periarticular tissues, classically causing excruciating acute inflammation of the first metatarsophalangeal joint of the big toe (**Podagra**).
- **Albuminuria & Renal Failure:**
- Normal glomerular capillary filtration barriers prevent passage of high-molecular-weight plasma proteins. If **albumin appears in urine (albuminuria / proteinuria)**, it indicates glomerular membrane breakdown and approaching **Kidney failure**.
- **Metabolic Effects of Prolonged Fasting:**
- Prolonged fasting rapidly exhausts hepatic glycogen reserves, initiating extensive lipolysis, ketogenesis (formation of acetoacetate and $\beta$-hydroxybutyrate), and protein catabolism via gluconeogenesis, placing severe metabolic burdens on the **Kidneys and Liver**.
- **Bulimia Nervosa:**
- Psychological eating disorder characterized by recurrent episodes of uncontrolled binge eating (patient is unable to stop eating) followed by compensatory purging (induced vomiting, laxatives). Distinct from **Anorexia Nervosa** (obsessive self-starvation due to distorted body image).
- **Anthophobia:** Extreme, abnormal, irrational psychological fear of **flowers**.

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## 3.4 Medical Pharmacology, Radioisotopes & Landmark Therapeutics

### Antibiotics, Sulpha Drugs & Analgesics

- **Coining of the Term "Antibiotic":**
- Coined in **1942** by American microbiologist **Selman Waksman**, who discovered **Streptomycin** (the first effective antibiotic against *Mycobacterium tuberculosis*, isolated from soil actinomycete *Streptomyces griseus*; awarded the Nobel Prize in 1952).
- **Sulpha Drugs (Sulfonamides):**
- Synthetic **antibacterial** agents (e.g., **Cibazole / Sulfathiazole**, Sulfanilamide) that act as competitive structural analogues of PABA (para-aminobenzoic acid), inhibiting bacterial dihydropteroate synthase and halting folic acid synthesis. Discovered by Gerhard Domagk (Prontosil, Nobel Prize 1939).

- **Broad-Spectrum Antibiotics:**
- Antibiotics active against both Gram-positive and Gram-negative bacteria (e.g., **Chloramphenicol**, Tetracycline, Amoxicillin). Chloramphenicol is used clinically for typhoid, bacterial meningitis, and rickettsial infections.
- **Aspirin (Acetylsalicylic Acid):**
- Non-steroidal, non-narcotic analgesic, antipyretic, and antiplatelet drug; inhibits cyclooxygenase (COX-1 and COX-2) enzymes. *Exam Trap*: Aspirin does **NOT** belong to narcotic analgesics (morphine, codeine, pethidine are narcotics).
- **Atropine:**
- Anticholinergic tropane alkaloid extracted from the deadly nightshade plant ***Atropa belladonna***; blocks muscarinic acetylcholine receptors, used clinically by ophthalmologists to **dilate the pupils (mydriasis)** and paralyze accommodation for retinal exams.
- **Pyrethrin:**
- Natural organic insecticide extracted from the dried flower seed heads of ***Chrysanthemum cinerariifolium*** (dalmatian chrysanthemum); neurotoxic to mosquitoes and insects, widely incorporated into household mosquito coils and liquid vaporizers.

### Radioisotopes in Clinical Medicine & Oncology

Radioactive isotopes are routinely used for targeted diagnostic imaging and radiotherapy:

| Radioisotope | Symbol | Clinical Application & Target Condition |
|---|---|---|
| **Radio-Sodium** | $^{24}Na$ | Measures the velocity of blood circulation and detects vascular occlusions, thrombosis, and arterial blockages. |
| **Phosphorus-32** | $^{32}P$ | Targeted radiotherapy for **Leukemia (blood cancer)** and polycythemia vera (abnormal proliferation of RBCs). |
| **Iodine-131** | $^{131}I$ | Diagnostic thyroid scanning and radioablation of hyperthyroidism, goitre, and **Thyroid Carcinoma**. |
| **Cobalt-60** | $^{60}Co$ | High-energy gamma radiation source utilized in external beam teletherapy for deep-seated malignant solid tumors. |
| **Arsenic-74** | $^{74}As$ | Diagnostic imaging and precise localization of **brain tumors**. |
| **Gold-198** | $^{198}Au$ | Interstitial radiotherapy for localized prostate and liver malignancies. |

### Cancer Therapeutics & Clinical Oncology

- **Platinum Chemotherapy Compounds:**
- **Platinum ($Pt$)**-based coordination complexes, notably **Cisplatin** (*cis*-diamminedichloroplatinum(II)) and Carboplatin, are widely used chemotherapy agents that bind to DNA, forming intrastrand crosslinks that trigger apoptosis in carcinomas (testicular, ovarian, bladder, and lung cancers).
- **Pap Smear (Papanicolaou Test):**
- Universal screening cytological test used for the early clinical detection of premalignant dysplasia and carcinoma of the **Uterine Cervix**.
- **Intensity-Modulated Radiotherapy (IMRT):**

- Advanced 3D conformal radiation oncology technique utilizing computer-controlled multi-leaf collimators to deliver precise, variable radiation doses conforming tightly to the 3D contour of tumors while sparing adjacent critical organs.
- **Arvid Carlsson & Parkinson's Disease:**
- Swedish pharmacologist **Arvid Carlsson** won the **2000 Nobel Prize in Physiology or Medicine** for demonstrating that dopamine is a vital neurotransmitter in the brain's basal ganglia, which directly led to the development of **L-DOPA therapy** for the treatment of **Parkinson's disease**.
- **DRDO Herbal Therapeutics:**
- **Lukoskin:** A polyherbal drug formulation developed by the Defence Research and Development Organisation (**DRDO**) for the clinical management of vitiligo (leukoderma).

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## 3.5 Sensory Organs, Clinical Diagnostics & Surgical Procedures

### Ophthalmology & Vision Correction

- **Myopia (Near-Sightedness):**
- A refractive defect where an individual can see **nearby objects clearly**, while distant objects appear blurred.
- The eyeball is too long or the corneal curvature is too steep; parallel light rays from distant objects converge and focus **in front of the retina**.
- Corrected using a **Concave lens** (diverging lens) with negative optical power.
- **Hypermetropia (Far-Sightedness):**
- Individual can see distant objects clearly, but near vision is blurred. Light rays focus **behind the retina**.
- Corrected using a **Convex lens** (converging lens) with positive optical power.
- **Astigmatism:**
- A refractive error caused by an irregular, asymmetric curvature of the cornea or crystalline lens (resembling a rugby ball rather than a sphere), causing light rays to focus at multiple points.
- Corrected using a **Cylindrical lens**.
- **Presbyopia ("Old-age vision"):**
- Gradual age-related loss of the lens's accommodation elasticity and ciliary muscle tone; corrected using **Bifocal lenses** (upper concave for distance, lower convex for reading).
- **Embryological Origin of the Retina:**
- The neural retina develops as an anatomical neuroectodermal evagination of the **Diencephalon** (forebrain optic vesicle).

### Emergency Medicine & Surgical Interventions

- **CPR (Cardiopulmonary Resuscitation):**
- An emergency lifesaving resuscitation procedure combining chest compressions ( $100–120\ \text{compressions}/\text{min}$) and rescue breathing to maintain vital oxygenated blood flow to the brain and heart during sudden cardiac arrest.
- **Coronary Artery Bypass Graft (CABG):**

- Surgical revascularization used in severe coronary artery disease to bypass blocked coronary arteries:
  - **Great Saphenous Vein** harvested from the **leg**.
  - **Internal Mammary (Thoracic) Artery** harvested from the **chest**.
  - **Radial Artery** harvested from the **forearm**.
- **Cath Lab (Catheterization Laboratory):**
- Specialized hospital interventional diagnostic examination room located in the **Cardiology department**, equipped with fluoroscopy imaging to perform coronary angiograms, balloon angioplasty, and cardiac stent implants.
- **Dialysis (Hemodialysis):**
- Clinical extracorporeal procedure used when there is severe dysfunction or acute/chronic failure of the **Kidneys**. Blood is circulated through an artificial dialyzer membrane to remove urea, creatinine, and excess potassium.
- **Blood Glucose Metrics:**
- Blood glucose concentration is clinically quantified in **milligrams per decilitre (mg/dL)**:
  - Normal Fasting Plasma Glucose: $70–99\ \mathrm{mg/dL}$
  - Impaired Fasting Glucose (Prediabetes): $100–125\ \mathrm{mg/dL}$
  - Diagnostic Threshold for Diabetes Mellitus: $\geq 126\ \mathrm{mg/dL}$ (or random $\geq 200\ \mathrm{mg/dL}$; $HbA_{1c} \geq 6.5\%$).
- Diabetes mellitus is caused by functional endocrine failure of the **Pancreas** (specifically insulin deficiency from $\beta$-cells of the Islets of Langerhans, discovered by **Frederick Banting and Charles Best, 1921**).
- **BCG Vaccine Administration Timing:**
- The BCG (*Bacillus Calmette-Guérin*) vaccine against tuberculosis must be administered **at birth or as soon as possible after birth (within the first few days/weeks)**.
- **Infrared Radiotherapy:**
- Invisible infrared electromagnetic radiation ($700\ \mathrm{nm} - 1\ \mathrm{mm}$) produces deep thermal penetration in muscular tissues, used in physical medicine for the relief of muscle spasms, stiffness, and chronic lumbago.

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| English Term | Hindi Term | Key Anchor |
|---|---|---|
| **Nutrition** (पोषण) | पोषण | Process of obtaining and using food for growth and energy |
| **Carbohydrate** (कार्बोहाइड्रेट) | कार्बोहाइड्रेट | Primary energy source; glucose, starch, cellulose; C:H:O = 1:2:1 |
| **Protein** (प्रोटीन) | प्रोटीन | Built from amino acids; body-building nutrient; enzymes are proteins |
| **Fat / Lipid** (वसा) | वसा / लिपिड | Energy reserve; fat-soluble vitamins ADEK; 9 kcal/g |
| **Vitamin** (विटामिन) | विटामिन | Organic micronutrient; essential in small amounts |
| **Mineral** (खनिज) | खनिज | Inorganic micronutrient; e.g., calcium, iron, iodine |

| English Term | Hindi Term | Key Anchor |
|---|---|---|
| **Enzyme** (एंजाइम) | एन्जाइम | Biological catalyst; protein in nature; specific action |
| **Deficiency Disease** (अपूर्णता रोग) | न्यूनता रोग | Disease caused by lack of specific nutrient |
| **Scurvy** (स्कर्वी) | स्कर्वी | Vitamin C deficiency; bleeding gums; treated by lemon/amla |
| **Rickets** (रिकेट्स) | सूखा रोग | Vitamin D deficiency (children); soft, bent bones |
| **Beriberi** (बेरीबेरी) | बेरी-बेरी | Vitamin B1 (Thiamine) deficiency; neurological + cardiovascular |
| **Night Blindness** (रतौंधी) | रतौंधी | Vitamin A deficiency; cannot see in dim light |
| **Anaemia** (रक्ताल्पता) | रक्ताल्पता | Iron deficiency; low haemoglobin; pale skin, fatigue |
| **Goitre** (घेंघा) | घेंघा | Iodine deficiency; enlarged thyroid gland |
| **Kwashiorkor** (क्वाशियोरकोर) | क्वाशियोरकोर | Protein deficiency; swollen belly in children |
| **Marasmus** (मरास्मस) | मरास्मस | Protein + calorie deficiency; severe wasting in children |
| **Pathogen** (रोगजनक) | रोगाणु / रोगजनक | Disease-causing organism: bacteria, virus, fungi, protozoa |
| **Antibiotic** (प्रतिजैविक) | प्रतिजैविक | Drug that kills/inhibits bacteria; Alexander Fleming (Penicillin, 1928) |
| **Vaccine** (टीका) | टीका | Preparation of killed/weakened pathogen; stimulates immunity |
| **Epidemic** (महामारी) | महामारी | Disease affecting many people in a community at the same time |
| **Pandemic** (वैश्विक महामारी) | वैश्विक महामारी | Worldwide epidemic; e.g., COVID-19, Spanish Flu |
| **Zoonosis** (जूनोसिस) | जूनोटिक रोग | Disease transmitted from animals to humans; rabies, plague, bird flu |
| **Vector** (वाहक) | वाहक | Organism carrying pathogen to host; mosquito, tick, sandfly |

## 3.2 Sugars, Carbohydrate Classes & Enzymes

- **Classification of Carbohydrates**:
- **Monosaccharides (Simple Sugars, $C_nH_{2n}O_n$)**:
  - **Glucose (Dextrose / Grape Sugar / Blood Sugar)**: Immediate cellular fuel; normal fasting physiological concentration is $70–100\ \mathrm{mg/dL}$; transported in bloodstream.
  - **Fructose (Levulose / Fruit Sugar)**: Ketohexose found in fruits and honey; **sweetest natural sugar**; unique fuel utilized by human spermatozoa (secreted by seminal vesicles).
  - **Galactose (Brain Sugar)**: Hydrolysed from milk lactose; converted to glucose in the liver.
- **Oligosaccharides (Disaccharides, $C_{12}H_{22}O_{11}$)**:

- **Sucrose (Table Sugar / Cane Sugar / Invert Sugar)**: Non-reducing sugar composed of $\alpha$-D-Glucose + $\beta$-D-Fructose linked by $\alpha$-1,2-glycosidic bond; hydrolysed by invertase/sucrase.
- **Lactose (Milk Sugar)**: Reducing sugar composed of $\beta$-D-Galactose + $\beta$-D-Glucose linked by $\beta$-1,4-glycosidic bond; least sweet natural sugar; deficiency of intestinal lactase causes **lactose intolerance** (bloating, cramps, osmotic diarrhoea).
- **Maltose (Malt Sugar)**: Reducing sugar composed of two $\alpha$-D-Glucose molecules linked by $\alpha$-1,4-glycosidic bond; product of starch hydrolysis by salivary and pancreatic amylase.

- **Polysaccharides (Complex Polymers, $(C_6H_{10}O_5)_n$)**:
  - **Starch**: Storage carbohydrate of plants; composed of unbranched water-soluble **Amylose** ($\alpha$-1,4 bonds) and branched insoluble **Amylopectin** ($\alpha$-1,4 and $\alpha$-1,6 bonds); stains deep **blue-black with iodine**.
  - **Glycogen (Animal Starch)**: Storage carbohydrate of animals and fungi; highly branched polymer of glucose; stains **red-violet with iodine**; stored primarily in liver and skeletal muscles.
  - **Cellulose**: Structural unbranched polysaccharide of plant cell walls composed of $\beta$-D-glucose linked by $\beta$-1,4-glycosidic bonds; provides dietary bulk and stimulates intestinal peristalsis; completely undigested by human enzymes.

- **Enzymes as Biological Catalysts**:
- Almost all enzymes are globular proteins (ribozymes like 23S rRNA are RNA exceptions); accelerate chemical reaction rates by several million-fold by **lowering the activation energy barrier** without undergoing permanent chemical alteration.
- Show exquisite substrate specificity (Fact and Key hypothesis of Emil Fischer; Induced Fit theory of Daniel Koshland).
- Many enzymes require non-protein chemical components called **Cofactors**: inorganic metal ions ( $Zn^{2+}$ for carbonic anhydrase, $Fe^{2+}$ for catalase, $Mg^{2+}$ for hexokinase) or organic **Coenzymes** (derived predominantly from B-complex vitamins, e.g. TPP from B1, FAD from B2, NAD from B3, CoA from B5).

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## 3.3 The Vitamin Spectrum: Fat-Soluble vs Water-Soluble

- **Vitamins: General Principles**:
- Vital organic micronutrients required in trace quantities for physiological regulation; do not yield direct metabolic energy.
- Term "Vitamine" coined by Polish biochemist **Casimir Funk (1912)** (originally believed to be "vital amines").
- The human body **cannot synthesise most vitamins** (must be obtained from diet); exceptions: **Vitamin D** (made in sunlit skin), **Vitamin K and B12** (synthesised by intestinal microflora in colon), and small amounts of **Niacin** (from tryptophan).
- **Fat-Soluble Vitamins (A, D, E, K)**:
- **Vitamin A (Retinol)**:
  - Converted from dietary provitamin carotenoids ($\beta$-carotene) in the intestinal mucosa and liver.
  - Forms **Rhodopsin (visual purple)** in retinal rod cells, essential for scotopic dim-light vision.

    - Deficiency cascade: **Nyctalopia (Night Blindness)** $\rightarrow$ **Xerophthalmia** (conjunctival dryness, Bitot's triangular foamy spots on sclera) $\rightarrow$ **Keratomalacia** (corneal softening, ulceration, permanent blindness) and hyperkeratosis of skin (*toad skin*).

- **Vitamin D (Calciferol)**:
    - Precursor 7-dehydrocholesterol in skin is converted to Cholecalciferol ($D_3$) by UV-B sunlight; converted in liver to 25-hydroxycalciferol, and activated in kidney by 1-$\alpha$-hydroxylase to **1,25-dihydroxycholecalciferol (Calcitriol)**.
    - Stimulates active intestinal absorption of $Ca^{2+}$ and phosphate, maintaining bone mineralization.
    - Deficiency in children: **Rickets** (failure of bone calcification, bow legs / genu varum, knock knees, rickety rosary on ribs, delayed closure of fontanelles).
    - Deficiency in adults: **Osteomalacia** (softening and demineralization of adult bones, severe bone pain and fractures).

- **Vitamin E (Tocopherol)**:
    - Group of tocopherols and tocotrienols ($\alpha$-tocopherol is most biologically active).
    - Major lipid-soluble antioxidant in biological membranes; scavenges reactive oxygen species (ROS) and terminates lipid peroxidation chain reactions; protects RBC membranes from fragility.
    - Richest sources: Wheat-germ oil, sunflower oil, safflower oil, nuts, and green vegetables.

- **Vitamin K (Phylloquinone / Menaquinone)**:
    - Post-translational cofactor for $\gamma$-glutamyl carboxylase in the liver, enabling calcium-binding activation of blood clotting factors **II (Prothrombin), VII, IX, and X**, as well as bone osteocalcin.
    - Deficiency: Defective blood coagulation, prolonged prothrombin time, excessive subcutaneous bruising, and uncontrollable haemorrhage.
    - Haemorrhagic disease of the newborn occurs because neonates have a sterile gut lacking bacterial flora and breast milk is low in Vitamin K (routine intramuscular Vitamin K injection given at birth).

- **Water-Soluble Vitamins (B-Complex & C)**:
- **Vitamin B1 (Thiamine)**:
    - Coenzyme TPP decarboxylates pyruvate to acetyl-CoA; critical for nervous system glucose oxidation.
    - Deficiency: **Beriberi**:
    - **Dry Beriberi**: Severe peripheral polyneuropathy, symmetrical foot-drop and wrist-drop, muscle wasting.
    - **Wet Beriberi**: Cardiovascular involvement, high-output congestive heart failure, peripheral pitting oedema.
    - **Wernicke-Korsakoff Syndrome**: Encephalopathy, ataxia, confusion, and memory loss in chronic alcoholics.
    - **Logic: Polished rice diets strip thiamine, causing beriberi (UPPCS 2019).**

- **Vitamin B2 (Riboflavin)**:
    - Essential for FMN and FAD coenzymes in cellular respiration.

    - Deficiency: Ariboflavinosis causing **Cheilosis** (cracking at lips), **Angular Stomatitis** (fissures at angles of mouth), and **Glossitis** (swollen magenta tongue).

- **Vitamin B3 (Niacin / Nicotinic Acid)**:
    - Precursor of NAD and NADP; deficiency causes **Pellagra** (characterised by 4 Ds: Dermatitis, Diarrhoea, Dementia, Death).
    - Casal's necklace is a pathognomonic broad dermatitis collar around the neck in pellagra patients exposed to sunlight.
- **Vitamin B6 (Pyridoxine)**:
    - Coenzyme for transaminases and amino acid metabolism; deficiency causes peripheral neuritis and microcytic anaemia. Drug **Isoniazid (INH)** used in TB therapy forms a complex with pyridoxine, requiring co-administration of Vitamin B6.
- **Vitamin B9 (Folic Acid)**:
    - Transfers one-carbon units in DNA synthesis; deficiency causes **Megaloblastic Anaemia** and fetal **Neural Tube Defects (Spina bifida)**.
- **Vitamin B12 (Cyanocobalamin)**:
    - Possesses a corrin ring coordinated to a central **Cobalt** atom; essential for normal erythropoiesis and myelin sheath maintenance.
    - Requires **Castle's Intrinsic Factor** (secreted by gastric parietal/oxyntic cells) for absorption in the ileum.
    - Deficiency: **Pernicious Anaemia** (autoimmune destruction of parietal cells, giant megaloblasts) and subacute combined degeneration of the spinal cord (tingling, ataxia).
- **Vitamin C (Ascorbic Acid)**:
    - Powerful water-soluble antioxidant; keeps prolyl and lysyl hydroxylase enzymes in reduced state for **collagen triple-helix cross-linking**; promotes intestinal non-haem iron absorption by reducing $Fe^{3+} \rightarrow Fe^{2+}$.
    - Deficiency: **Scurvy** (breakdown of capillaries, bleeding swollen spongy gums, loss of teeth, petechiae, delayed wound healing, anaemia).
    - Extremely thermolabile: destroyed by heating, boiling, and prolonged drying. Amla is the richest Indian food source.

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## 3.4 Mineral Dynamics, Toxic Heavy Metals & Environmental Poisonings

- **Macro-minerals & Micro-minerals**:
- **Macro-minerals ($> 100\ \text{mg/day}$)**: Calcium, Phosphorus, Potassium, Sulfur, Sodium, Chloride, Magnesium.
- **Micro-minerals / Trace Elements ($< 100\ \text{mg/day}$)**: Iron, Zinc, Copper, Iodine, Fluoride, Manganese, Selenium, Molybdenum, Chromium, Cobalt.
- **Iron ($Fe$)**: Core functional component of **Haemoglobin** in blood and **Myoglobin** in muscle; deficiency causes **Microcytic Hypochromic Anaemia** (fatigue, pallor, spoon-shaped nails / **Koilonychia**); rich sources: Jaggery (gur), spinach, liver, red meat, legumes.

- **Iodine ($I$)**: Essential constituent of thyroid hormones **Thyroxine ($T_4$) and Triiodothyronine ($T_3$)**; deficiency causes simple endemic **Goitre** (thyroid enlargement), **Cretinism** in infants (stunted growth, mental retardation, deaf-mutism), and **Myxedema** in adults; prevented by iodized salt (fortified with potassium iodate $KIO_3$).
- **Zinc ($Zn$)**: Component of $> 300$ enzymes (carbonic anhydrase, alcohol dehydrogenase, DNA/RNA polymerases); essential for insulin storage, taste acuity (**Hypogeusia**), spermatogenesis, and wound healing; deficiency causes growth retardation, delayed sexual maturation, and **Acrodermatitis enteropathica**.
- **Environmental Toxic Heavy Metal Poisonings (Direct PSC Common Traps)**:

| Toxic Agent / Contaminant | Environmental Source | Disease / Clinical Syndrome | Pathological Manifestation |
|---|---|---|---|
| **Methyl Mercury ($Hg$)** | Industrial effluent in fish/shellfish | **Minamata Disease** | Severe neurotoxicity, sensory ataxia, constriction of visual fields, hearing loss, convulsions |
| **Cadmium ($Cd$)** | Mining runoff in river irrigation | **Itai-Itai Disease ("Ouch-Ouch")** | Painful osteomalacia, multiple pseudofractures, renal tubular proteinuria |
| **Lead ($Pb$)** | Leaded petrol, old paints, pipes | **Plumbism / Saturnism** | Microcytic anaemia, basophilic stippling of RBCs, **Burtonian Blue Line** on gums, wrist drop, encephalopathy |
| **Nitrate ($NO_3^-$)** | Fertilizer runoff in well drinking water | **Blue Baby Syndrome (Methemoglobinemia)** | Nitrate reduced to nitrite; forms methemoglobin; severe cyanosis, asphyxia, blue-tinged skin |
| **Arsenic ($As$)** | Groundwater in alluvial aquifers | **Blackfoot Disease**, Hyperkeratosis | Peripheral vascular gangrene of feet/toes, cutaneous hyperkeratosis, bladder/lung cancer |
| **Excess Fluoride ($F^-$)** | Geogenic groundwater $> 1.5–2$ ppm | **Dental & Skeletal Fluorosis** | Brown mottled dental enamel; bone deformities, stiff joints, **Knock-knee syndrome (Genu valgum)** |
| **Asbestos Fibres** | Asbestos mining, roofing, insulation | **Asbestosis & Mesothelioma** | Diffuse pulmonary fibrosis; malignant cancer of the pleura/peritoneum |
| **Silica Dust ($SiO_2$)** | Stone quarrying, sandblasting, pottery | **Silicosis** | Chronic fibrotic nodular lung disease; pre-disposes to active pulmonary tuberculosis |
| **Coal Dust** | Underground coal mining | **Anthracosis ("Black Lung")** | Carbon accumulation in pulmonary macrophages, progressive massive fibrosis |
| **Cotton Dust** | Textile mill spinning and carding | **Byssinosis ("Brown Lung")** | Monday chest tightness, bronchospasm, chronic bronchitis |
| **Sugar Cane Bagasse** | Sugar and paper manufacture | **Bagassosis** | Hypersensitivity pneumonitis from thermophilic actinomycetes |

## 3.5 Infectious, Communicable & Vector-Borne Diseases

- **Modes of Transmission & Disease Terminology**:
- **Infectious / Communicable**: Diseases caused by biological pathogens capable of being transmitted from an infected host to a susceptible individual via air, water, food, vector, or direct physical contact.
- **Vector**: A living organism (usually an arthropod) that transfers an infectious agent from one host to another (e.g. mosquito, tick, sandfly, flea).
- **Vehicle**: A non-living contaminated intermediate substance transmitting pathogens (e.g. water, milk, food, surgical instruments).
- **Endemic**: Constant presence of a disease within a geographic area (e.g. goitre in sub-Himalayan belt).
- **Epidemic**: Sudden outbreak of disease attacking many people simultaneously in a community.
- **Pandemic**: Worldwide epidemic spreading across continents (e.g. Spanish Flu 1918, COVID-19 2020).
- **Major Bacterial Diseases of Public Health Significance**:
- **Tuberculosis**: Sputum smear microscopy (Ziehl-Neelsen acid-fast staining), GeneXpert (rapid molecular testing), Mantoux test ($> 10\ \mathrm{mm}$ induration = positive). DOTS uses daily fixed-dose combination therapy.
- **Typhoid Fever**: Widal test measures agglutinating antibody titres against $O$ (somatic) and $H$ (flagellar) antigens of *Salmonella typhi*.
- **Cholera**: Massive fluid loss ($10\text{–}20\ \mathrm{litres/day}$); death results from hypovolemic circulatory shock; treatment of choice is immediate aggressive hydration with WHO-formulated ORS (contains NaCl, KCl, Trisodium citrate, and anhydrous Glucose).
- **Diphtheria, Pertussis, Tetanus**: Covered by trivalent **DPT / Pentavalent Vaccine** administered under the Universal Immunisation Programme (UIP).
- **Major Viral Diseases of Public Health Significance**:
- **Dengue Fever**: Caused by Flavivirus; transmitted by female *Aedes aegypti* (breeds in clean domestic stagnant water; day-biter). Tourniquet test indicates capillary leakage; marked by precipitous decline in platelet counts ($< 100,000/\mu\mathrm{L}$); critical management requires fluid monitoring, avoiding NSAIDs like aspirin/ibuprofen (which exacerbate bleeding).
- **Chikungunya & Zika**: Also transmitted by *Aedes aegypti*; Zika virus infection in pregnant women causes severe congenital fetal **Microcephaly** (abnormally small head/brain).
- **Nipah Virus (NiV)**: Enveloped negative-sense single-stranded RNA Henipavirus (family Paramyxoviridae); natural reservoir host is the **Fruit Bat / Flying Fox (genus *Pteropus*)**; transmission occurs via consumption of raw date palm sap or fruits bitten by infected bats, contact with intermediate infected domestic pigs, or direct human-to-human droplet contact; clinical hallmark: acute encephalitis, fever, headache, altered consciousness, and rapid progression to coma ($40\text{–}75\%$ case fatality rate); Kerala outbreaks: Kozhikode (2018), Malappuram (2024), Palakkad (2025).
- **COVID-19 (SARS-CoV-2)**: Single-stranded positive-sense enveloped RNA coronavirus; binds to **ACE2 (Angiotensin-Converting Enzyme 2) receptors** via its trimeric surface **Spike (S) glycoprotein**;

diagnosed by **RT-PCR** (gold standard detecting viral $E$, $N$, and $RdRp$ genes) or Rapid Antigen Test (RAT).

- **Major Protozoan & Helminthic Diseases**:
- **Malaria**: Four species of *Plasmodium*: *P. vivax* (benign tertian, 48-hr cycle), *P. falciparum* (malignant tertian, irregular/48-hr cycle, cerebral malaria), *P. malariae* (quartan, 72-hr cycle), *P. ovale* (ovale tertian). Asexual erythrocytic schizogony causes synchronous bursting of RBCs, dumping **haemozoin** toxin into plasma which triggers cyclic chills and fever. Biological control: Gambusia fish (*Gambusia affinis*), guppy fish, and dragonfly nymphs feed voraciously on mosquito larvae in ponds.
- **Filariasis (Elephantiasis)**: Caused by *Wuchereria bancrofti* and *Brugia malayi*; transmitted by nocturnal female *Culex* mosquitoes; adult worms lodge in human **inguinal and scrotal lymphatics**, causing chronic lymphangitis, gross obstruction, and elephantine lymphedema of legs and scrotum (**Hydrocele**); treated with Diethylcarbamazine (DEC) and Albendazole.

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### Master Bacterial Pathogens, Clinical Manifestations & Antibiotics

| Bacterial Disease | Pathogen / Morphology | Primary Affected Organs | Transmission & Hallmarks | Treatment & Vaccines |
|---|---|---|---|---|
| **Cholera** | *Vibrio cholerae* (Gram-negative comma-shaped bacillus) | Small intestine | Contaminated water & food; severe watery "rice-water" diarrhoea, acute dehydration | Oral Rehydration Solution (ORS), Azithromycin, Tetracycline |
| **Plague ("Black Death")** | *Yersinia pestis* (*Pasteurella pestis*) (Gram-negative rod) | Lymphatic nodes (buboes in groin, neck, axilla) | Vector: **Rat flea (*Xenopsylla cheopis*)** from wild rodents; bubonic, septicemic, and pneumonic (fatal within 24 hrs) | Streptomycin, Gentamicin, Doxycycline |
| **Tuberculosis (TB)** | *Mycobacterium tuberculosis* (Acid-fast rod) | Lungs primarily; pleura, lymph nodes, spine (Pott's disease) | Airborne droplets; fever, night sweats, hemoptysis, chronic cough; **#1 infectious killer worldwide (~1.25M annual deaths)** | **BCG vaccine** at birth; **Mantoux Tuberculin Skin Test (TST)**; DOTS (Isoniazid, Rifampicin, Pyrazinamide, Ethambutol, PAS) |
| **Leprosy (Hansen's Disease)** | *Mycobacterium leprae* & *M. lepromatosis* (Acid-fast bacilli) | Skin, peripheral nerves, upper respiratory mucosa | Nasal droplets; skin hypo-pigmented patches, sensory loss, nerve thickening, deformities | Multi-Drug Therapy (MDT: Dapsone, Rifampicin, Clofazimine, PAS) |
| **Pneumonia** | *Streptococcus pneumoniae* & *Haemophilus influenzae* | Alveolar air sacs of lungs | Droplets; alveoli fill with fluid/exudate, chest pain, dyspnea, fever | **Pneumococcal Conjugate Vaccine (PCV)**; Penicillin, Ceftriaxone |

| Bacterial Disease | Pathogen / Morphology | Primary Affected Organs | Transmission & Hallmarks | Treatment & Vaccines |
|---|---|---|---|---|
| **Typhoid (Enteric Fever)** | *Salmonella typhi* & *S. paratyphi* (Gram-negative flagellated bacillus) | Small intestine (Peyer's patches), bloodstream | Fecal-oral route via food/water; sustained high stepped fever, headache, "pea-soup" diarrhoea, intestinal perforation | **Widal Agglutination Test**; Ceftriaxone, Ciprofloxacin, Chloramphenicol |
| **Diphtheria** | *Corynebacterium diphtheriae* (Gram-positive rod) | Throat, upper respiratory tract | Droplet infection; gray fibrous pseudomembrane covering tonsils/pharynx, suffocating airway obstruction | **DPT Vaccine (Toxoid)**; Diphtheria antitoxin + Erythromycin |
| **Pertussis (Whooping Cough)** | *Bordetella pertussis* (*Haemophilus pertussis*) | Tracheobronchial respiratory mucosa | Airborne droplets; paroxysmal violent hacking cough spells ending in high-pitched inspiratory "whoop" | **DPT Vaccine**; Azithromycin |
| **Tetanus ("Lockjaw")** | *Clostridium tetani* (Anaerobic gram-positive spore-forming rod) | Central nervous system motor pathways | Spores enter via deep contaminated wounds/burns; releases **tetanospasmin** neurotoxin causing severe involuntary muscle spasms, trismus (jaw fact), opisthotonos | **DPT / Tetanus Toxoid (TT) / ATS (Anti-Tetanus Serum)**; Muscle relaxants |
| **Syphilis** | *Treponema pallidum* (Spirochaete bacterium) | Genital tract, skin, cardiovascular & CNS | Direct sexual contact, transplacental; primary painless hard ulcer (**chancre**), secondary rash, tertiary gummas | **Penicillin G**; VDRL / RPR serological tests |
| **Gonorrhoea** | *Neisseria gonorrhoeae* (Gram-negative diplococcus) | Genitourinary mucous membranes | Sexual transmission; painful micturition, purulent urethral discharge; can cause pelvic inflammatory disease (PID) | Ceftriaxone + Azithromycin |
| **Anthrax ("Splenic Fever")** | *Bacillus anthracis* (Spore-forming Gram-positive rod) | Spleen (marked splenomegaly), skin, lungs, intestines | Zoonotic: contact with infected livestock, hides, wool ("wool-sorter's disease"); weapon of bioterrorism | Ciprofloxacin, Doxycycline |

| Bacterial Disease | Pathogen / Morphology | Primary Affected Organs | Transmission & Hallmarks | Treatment & Vaccines |
|---|---|---|---|---|
| **Botulism (Food Intoxication)** | *Clostridium botulinum* (Anaerobic spore-forming bacterium) | Peripheral neuromuscular junctions | **Food-borne intoxication**: Ingestion of preformed lethal neurotoxin in improperly canned/stored alkaline foods (low oxygen); blocks acetylcholine release → flaccid paralysis, respiratory failure | Polyvalent botulinum antitoxin; mechanical ventilation |

- **Antimicrobial Pharmacology & Landmark Discoveries**:
- **Selman Waksman (1942)**: Coined the term **"Antibiotic"** to designate biochemical substances produced by microorganisms that inhibit or destroy competing microbes. Discovered **Streptomycin** from *Streptomyces griseus* (effective against TB; Nobel Prize 1952).
- **Alexander Fleming (1928)**: Discovered **Penicillin**, the first natural antibiotic, from the green mould *Penicillium notatum*, revolutionizing bacterial therapeutics.
- **Sulpha Drugs (Sulphonamides)**: First synthetic chemical antibacterial agents containing the sulphanilamide nucleus. **Prontosil** discovered by German physician **Gerhard Domagk (1932)** (Nobel Prize 1939). Synthetic sulpha derivatives include **Cibazole, Sulphadiazine, Sulphaguanidine**.
- **Broad-Spectrum vs Narrow-Spectrum Antibiotics**: Broad-spectrum agents act against a wide variety of both Gram-positive and Gram-negative bacteria (e.g. **Ampicillin, Chloramphenicol / Chloronitromycin, Tetracyclines**). Chloramphenicol is used for bacterial meningitis, typhoid, plague, and ocular conjunctivitis ointments.
- **Antimicrobial Resistance (AMR)**: Multi-drug resistance (MDR) in pathogenic bacteria is driven primarily by excessive and incorrect dosing of antibiotics, over-the-counter self-medication, non-therapeutic antibiotic growth promoters in commercial poultry/livestock feeds, and pharmaceutical wastewater discharge into river basins.

### Master Fungal Pathogens & Clinical Mycology

- **Mycology Fundamentals**: Fungi are eukaryotic, heterotrophic (saprophytic or parasitic) organisms possessing a cell wall made of **Chitin** (polymer of N-acetylglucosamine). Vegetative body consists of filamentous **hyphae** forming an interwoven **mycelium**.
- **Landmark Fungal Infections & Mycotoxicoses**:
- **Aflatoxins**: Potent hepatotoxic, mutagenic, and carcinogenic mycotoxins synthesized by moulds **Aspergillus flavus** and *Aspergillus parasiticus* infesting poorly stored agricultural commodities (peanuts, groundnuts, maize, cottonseed). Aflatoxin $B_1$ induces DNA mutations leading directly to **hepatocellular carcinoma (liver cancer)**.
- **Ergotism ("Saint Anthony's Fire" / Ergotoxicosis)**: Toxicosis caused by ingestion of alkaloids (ergotamine) produced by **Claviceps purpurea** infecting rye, barley, and cereal grains. Induces severe peripheral vasoconstriction resulting in burning sensations, gangrene of fingers and toes, hallucinations, and convulsions.

- **Athlete's Foot (Tinea pedis)**: Contagious superficial fungal infection of the interdigital skin of the toes; caused by dermatophyte fungi **Trichophyton, Epidermophyton, and Microsporum**. Symptoms include erythema, severe pruritus, skin maceration, and painful fissures.
- **Mucormycosis ("Black Fungus")**: Life-threatening opportunistic infection caused by *Mucormycetes* moulds (e.g. *Rhizopus, Mucor*). Prevalent in immunocompromised patients, particularly poorly controlled diabetics and post-COVID-19 patients treated with heavy systemic corticosteroids. Affects the rhino-orbito-cerebral tracts causing facial swelling, ptosis, proptosis, black necrotic eschar, blindness, and fatal cerebral invasion. Treated via intravenous **Amphotericin B** and surgical debridement.
- **Fungal Alopecia & Dermatitis**: Dermatophytic tinea capitis causing patch baldness; Seborrheic dermatitis (inflammatory scaling of scalp) caused by *Malassezia* yeasts.
- **Mycetism (Poisonous Mushrooms)**: Ingestion of wild toxic mushrooms such as the Death Cap (*Amanita phalloides*); contains the fatal cyclic octapeptide **$\alpha$-amanitin**, which irreversibly inhibits RNA polymerase II, precipitating fulminant hepatic and renal failure (causes 95% of fatal mushroom poisonings worldwide).

### Master Parasitic & Helminthic Pathogens (Protozoa, Flatworms & Roundworms)

| Parasitic Disease | Causative Parasite & Taxonomy | Intermediate / Biological Vector | Clinical Pathology & Cardinal Symptoms |
|---|---|---|---|
| **Malaria** | *Plasmodium* (*P. vivax, P. falciparum, P. malariae, P. ovale*) (Protozoa, Sporozoa) | Female **Anopheles mosquito** (Definitive Host); Humans are Intermediate Host | Erythrocytic schizogony causes synchronous rupture of RBCs every 48–72 hours, releasing pyrogenic **hemozoin** toxin provoking shaking chills, spiking fever, drenching sweats, severe anemia, and **splenomegaly (enlarged spleen)**.<br>- *P. vivax*: Accounts for ~65% of Indian cases.<br>- *P. falciparum*: Causes lethal **Malignant / Cerebral Malaria** with microvascular plugging in brain capillaries. |
| **Kala-azar (Visceral Leishmaniasis / "Black Fever")** | *Leishmania donovani* (Flagellated Protozoa) | Female **Sandfly (*Phlebotomus argentipes*)** | Parasite invades reticuloendothelial system; marked hepatosplenomegaly, cachexia, pancytopenia (anemia, leukopenia, thrombocytopenia), and dark **greyish-black hyperpigmentation** of facial and extremity skin. Second largest parasitic killer after malaria. |
| **African Sleeping Sickness (Trypanosomiasis)** | *Trypanosoma brucei gambiense* & *T. b. rhodesiense* (Protozoa) | **Tse-tse fly (*Glossina* species)** | Painful chancre at bite site; parasite penetrates blood-brain barrier into CSF, inducing chronic meningoencephalitis, daytime somnolence, nocturnal insomnia, progressive coma, and death. |

| Parasitic Disease | Causative Parasite & Taxonomy | Intermediate / Biological Vector | Clinical Pathology & Cardinal Symptoms |
|---|---|---|---|
| **Amoebiasis (Amoebic Dysentery)** | *Entamoeba histolytica* (Protozoa, Sarcodina) | Fecal-oral route via cysts in contaminated water/food | Trophozoites ulcerate colon mucosa (flask-shaped ulcers), causing severe abdominal cramping and bloody mucoid diarrhea; can metastasize to liver forming amoebic liver abscesses. |
| **Schistosomiasis ("Snail Fever" / Bilharzia)** | *Schistosoma haematobium, S. mansoni* (Trematode / Flatworm) | Freshwater aquatic snails; free-swimming cercariae penetrate human skin | Worms dwell in mesenteric or vesical venous plexuses; hematuria (blood in urine), bladder calcification, chronic liver fibrosis. |
| **Fascioliasis (Liver Rot)** | *Fasciola hepatica* & *F. gigantica* (Liver Flukes, Trematodes) | Primary host: Sheep & cattle; Intermediate host: Snails; Vector: raw watercress | Adult flukes obstruct biliary ducts; jaundice, right upper quadrant pain, hepatomegaly. |
| **Taeniasis & Neurocysticercosis (NCC)** | *Taenia solium* (Pork Tapeworm, Cestode / Flatworm) | Ingestion of undercooked "measly" pork containing cysticerci or fecal-oral ingestion of tapeworm eggs | Ingested eggs hatch into oncospheres that burrow through intestinal wall and encyst in muscles, eyes, and the **brain (Neurocysticercosis)**; **leading cause of acquired adult-onset epilepsy and seizures in developing nations** (as famously suffered by tennis star Leander Paes). |
| **Lymphatic Filariasis (Elephantiasis)** | *Wuchereria bancrofti* & *Brugia malayi* (Nematodes / Roundworms) | Female **Culex mosquito** (*Culex quinquefasciatus*) | Adult filarial nematodes lodge in lymphatic vessels and nodes, provoking chronic obstructive inflammation and massive irreversible lymphedema, hypertrophy of lower limbs (**elephantiasis**), and scrotal swelling (**hydrocele**). |
| **Ascariasis** | *Ascaris lumbricoides* (Giant Intestinal Roundworm, Nematode) | Fecal-oral ingestion of embryonated eggs from contaminated soil/food | Larval pulmonary migration (Loeffler's syndrome with eosinophilia and cough); adult worms in small intestine provoke malnutrition, colicky abdominal pain, and intestinal lumen obstruction. Treated with Albendazole/Mebendazole. |

- **Sir Ronald Ross (1897)**: British IMS physician working in Secunderabad, India; discovered *Plasmodium* oocysts in the stomach wall of female *Anopheles* mosquitoes, confirming the mosquito vector of malaria. Awarded the Nobel Prize in Physiology or Medicine in 1902.
- **Malaria Chemoprophylaxis & Control**:
- Traditional natural drug: **Quinine** extracted from *Cinchona* bark.
- Synthetic antimalarials: **Chloroquine, Primaquine, Mefloquine, Atovaquone**.

- Modern frontline artemisinin-based therapies (ACT): **Artemisinin, Arteether** (derived from *Artemisia annua*).
- Biological Vector Control: Introduction of larvivorous surface-feeding fish **Gambusia affinis** (Mosquitofish) and ducks into freshwater ponds to devour mosquito larvae.
- Landmark Malaria Vaccines: **RTS,S/AS01 (Mosquirix)** (recombinant circumsporozoite protein vaccine) and **R21/Matrix-M** (developed by Oxford University and manufactured at scale by **Serum Institute of India, Pune**; received WHO recommendation in October 2023 for pediatric immunization in endemic regions).
- **World Malaria Day**: Observed annually worldwide on **25 April**.

## 3.6 Non-Communicable Diseases, Oncology & Metabolic Disorders

- **Classification of Non-Communicable Diseases (NCDs)**:
- Account for $> 60\%$ of total adult deaths in India; driven by four major behavioural risk factors: tobacco use, unhealthy diet, physical inactivity, and harmful alcohol consumption.
- Four major categories: Cardiovascular Diseases, Cancers, Chronic Respiratory Diseases (COPD, Asthma), and Diabetes.
- **Cancer Pathogenesis & Clinical Oncology**:
- Normal cells show **Contact Inhibition** (inhibition of cell division upon physical contact with neighbouring cells); cancer cells lose this property, dividing relentlessly into abnormal masses called **Neoplasms / Tumours**.
- **Benign Tumours**: Remain localized to original site; encapsulated; do not invade surrounding tissues or spread to distant sites.
- **Malignant Tumours**: Non-capsulated; invade and destroy surrounding tissues; shed neoplastic cells into blood/lymph channels (**Metastasis** — the most dreaded hallmark of cancer).
- **Carcinogens**: Agents that induce cancer:
  - Physical: UV rays, X-rays, gamma rays.
  - Chemical: Tobacco smoke (contains benzo[a]pyrene, causing lung and oral cancer), coal tar, asbestos, aniline dyes (bladder cancer).
  - Biological: Oncogenic viruses (HPV $\rightarrow$ cervical cancer; Hepatitis B & C $\rightarrow$ liver cancer; Epstein-Barr Virus / EBV $\rightarrow$ Burkitt's lymphoma).
- **Diagnostic Biomarkers & Techniques**:
  - **Biopsy**: Microscopic histopathological examination of excised tissue section.
  - **Pap Smear (Papanicolaou Test)**: Cytological screening test for cervical dysplasia and cancer.
  - **PSA (Prostate-Specific Antigen)**: Serum protein biomarker elevated in prostate adenocarcinoma.
  - **CA-125**: Serum biomarker for ovarian cancer.
  - **Alpha-Fetoprotein (AFP)**: Biomarker for hepatocellular carcinoma.
- **Diabetes Mellitus: Etiology & Clinical Management**:
- Endocrine disorder arising from defects in insulin secretion, insulin action, or both:

- **Physiological Action of Insulin**: Secreted by $\beta$-cells of pancreatic islets of Langerhans; promotes glucose uptake into skeletal muscle and adipose cells via GLUT-4 transporters; stimulates **Glycogenesis** (glucose $\rightarrow$ glycogen) in liver; inhibits **Glycogenolysis** and **Gluconeogenesis**; promotes lipogenesis and protein synthesis.
- **Glucagon**: Secreted by $\alpha$-cells; acts antagonistically to insulin to elevate blood glucose levels by triggering rapid glycogenolysis.
- **Clinical Triad of Diabetes**: **Polyuria** (frequent urination), **Polydipsia** (increased fluid intake), and **Polyphagia** (voracious appetite).
- **Long-term Microvascular Complications**: Diabetic Retinopathy (blindness), Diabetic Nephropathy (renal glomerulosclerosis), and Diabetic Neuropathy (loss of peripheral sensation $\rightarrow$ diabetic foot ulcers).
- **Glycated Haemoglobin (HbA1c)**: Glucose binds non-enzymatically to the N-terminal valine of haemoglobin $\beta$-chains; reflects ambient average blood glucose concentration over the **past 90–120 days** (erythrocyte lifespan); unaffected by day-to-day dietary fluctuations:
  - Normal: $< 5.7\%$
  - Pre-diabetes: $5.7\%$–$6.4\%$
  - **Diabetes**: $\geq \mathbf{6.5\%}$

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### Master Human Genetic Disorders: Chromosomal Aberrations & Mendelian Inheritances

| Genetic Disorder | Inheritance Pattern / Genetic Cause | Primary Biochemical / Cellular Defect | Cardinal Clinical Manifestations & Must-Score Facts |
|---|---|---|---|
| **Haemophilia ("Royal Disease / Bleeder's Disease")** | **X-Linked Recessive**; carried by asymptomatic heterozygous females ($X^H X^h$), manifests clinically in hemizygous males ($X^h Y$) | Deficiency of clotting Factor VIII (Haemophilia A) or Factor IX (Haemophilia B, Christmas disease) | Prolonged coagulation time; minor cuts cause persistent, life-threatening internal/external hemorrhage; passed down European royal houses from Queen Victoria. |
| **Color Blindness (Daltonism)** | **X-Linked Recessive**; gene located on X-chromosome | Genetic defect in red- or green-sensitive retinal cone photopigments | Inability to distinguish red from green hues; affected males pass mutated X exclusively to daughters (making them obligate carriers); sons inherit Y from father and thus cannot inherit trait from father. Screened via **Ishihara 38-Plate Test**. |

| Genetic Disorder | Inheritance Pattern / Genetic Cause | Primary Biochemical / Cellular Defect | Cardinal Clinical Manifestations & Must-Score Facts |
|---|---|---|---|
| **Phenylketonuria (PKU)** | **Autosomal Recessive**; inborn error of amino acid metabolism | Inherited deficiency of hepatic enzyme **Phenylalanine Hydroxylase** | Inability to convert phenylalanine to tyrosine leads to toxic accumulation of phenylalanine and phenylpyruvic acid in cerebrospinal fluid and blood; causes **severe progressive mental retardation / idiocy**, microcephaly, and hypopigmentation. |
| **Alkaptonuria (AKU)** | **Autosomal Recessive**; mutation in HGD gene | Defective homogentisate 1,2-dioxygenase in tyrosine metabolism pathway | Accumulation of homogentisic acid (alkapton); **urine turns dark black upon standing on exposure to air**; ochronosis (bluish-black pigmentation of cartilages and sclera) and early severe degenerative arthritis. |
| **Albinism** | **Autosomal Recessive**; enzyme mutation | Total congenital absence or defect of the copper-containing enzyme **Tyrosinase** | Complete failure to synthesize **Melanin** pigment; snowy white hair, milky pinkish-white skin, translucent pink irises; extreme photophobia, sunburn, and high risk of cutaneous carcinoma. |
| **Severe Combined Immunodeficiency (SCID / "Bubble Baby Disease")** | Autosomal recessive (ADA deficiency) or X-linked | Adenosine deaminase mutation or IL-2R $\gamma$-chain defect halting T and B cell maturation | Complete absence of adaptive cell-mediated and humoral immunity; infants die of minor infections unless isolated in sterile plastic isolator bubbles; curable via allogeneic bone marrow transplantation or retroviral ex-vivo gene therapy. |
| **Sickle Cell Anaemia (SCD)** | **Autosomal Recessive / Codominant**; point mutation in $HBB$ gene on chromosome 11 | Single nucleotide substitution ( $GAG \rightarrow GTG$) replacing glutamic acid with valine at codon 6 of $\beta$-globin chain | Deoxygenated mutant hemoglobin ($HbS$) polymerizes into rigid fibers, twisting RBCs into rigid crescent/sickle shapes; microvascular occlusion, painful vaso-occlusive bone crises, splenic infarction, chronic hemolytic anemia. **National Sickle Cell Elimination Mission** launched July 2023 in Shahdol (MP) to eliminate transmission by **2047**. |

| Genetic Disorder | Inheritance Pattern / Genetic Cause | Primary Biochemical / Cellular Defect | Cardinal Clinical Manifestations & Must-Score Facts |
|---|---|---|---|
| **Thalassemia** | **Autosomal Recessive** blood disorder | Quantitative deficiency in synthesis of $\alpha$-globin ($\alpha$-thalassemia) or $\beta$-globin ($\beta$-thalassemia) chains | Imbalanced globin chain synthesis leads to erythrocyte hemolysis, microcytic hypochromic anemia, skeletal deformities (iron overload from frequent blood transfusions), and hepatosplenomegaly. |
| **Duchenne Muscular Dystrophy (DMD)** | **X-Linked Recessive** | Mutation in the *dystrophin* gene on X-chromosome | Progressive necrosis and degeneration of skeletal muscle fibers; early childhood onset of proximal muscle weakness, calf pseudohypertrophy, wheelchair dependence by adolescence. |
| **Down Syndrome (Trisomy 21 / "Mongolian Idiocy")** | **Aneuploidy: Trisomy of Chromosome 21** ($2n+1 = 47$ chromosomes, $45A + XX/XY$) | Primary non-disjunction of chromosome 21 pair during maternal gametogenesis | Flat facial profile, epicanthic eye folds, protruding furrowed tongue, transverse single palmar crease (simian crease), congenital endocardial cushion heart defects, and mild-to-moderate intellectual disability. |
| **Turner Syndrome** | **Sex Chromosomal Monosomy** ($2n-1 = 45$ chromosomes, $44A + XO$) | Loss of one paternal or maternal X-chromosome in females | Phenotypically female; short stature, webbed neck (pterygium colli), shield-shaped chest, coarctation of aorta, **rudimentary streak ovaries causing absolute sterility (amenorrhea)**. |
| **Klinefelter Syndrome** | **Sex Chromosomal Trisomy in Males** ($2n+1 = 47$ chromosomes, $44A + XXY$) | Meiotic non-disjunction contributing extra X-chromosome to male zygote | Phenotypically male; tall eunuchoid body proportions, sparse facial/body hair, high-pitched voice, **gynecomastia (enlarged male breast tissue)**, micro-orchidism with azoospermia (sterility). |

### Oncology, Modern Radiation Medicine, Environmental Poisonings & Metabolic Pathology

- **Mechanisms of Carcinogenesis & Metastasis**:
- Cancer involves uncontrolled mitotic proliferation of transformed neoplastic cells, evasion of apoptosis, and induction of neo-angiogenesis.
- **Metastasis Hierarchy**: Primary tumour cells invade local tissue $\rightarrow$ intravasate into blood and lymphatic vessels $\rightarrow$ travel via circulation $\rightarrow$ arrest and extravasate across capillary endothelium into distant organs $\rightarrow$ form micrometastases.

- **Oncogenes**: Mutated, hyper-activated cellular proto-oncogenes encoding oncoproteins that drive continuous mitotic signaling. First discovered retroviral oncogene was the *src* gene of **Rous Sarcoma Virus** (causes connective tissue sarcomas in fowl).
- **Carcinogens**: Physical (UV radiation, ionizing gamma/X-rays), chemical (tobacco benzopyrenes, asbestos, arsenic, aromatic amines, aflatoxins), and biological oncoviruses (HPV, HBV, EBV).
- **Radioisotopes & Advanced Radiation Modalities in Oncology**:
- **Cobalt-60 ($^{60}\mathrm{Co}$)**: Synthetic radioisotope that emits high-energy penetrating **Gamma ($\gamma$) rays**; used in external beam teletherapy for deep solid tumours.
- **Phosphorus-32 ($^{32}\mathrm{P}$)**: Pure $\beta$-emitter; incorporates into rapidly dividing bone marrow cells; used to treat **Leukaemia (blood cancer)** and Polycythemia vera.
- **Iodine-131 ($^{131}\mathrm{I}$)**: Concentrated selectively by sodium-iodide symporters in thyroid follicle cells; used in diagnostic thyroid scans and treatment of thyroid carcinoma and thyrotoxicosis.
- **Arsenic-74 ($^{74}\mathrm{As}$)**: Positron/gamma emitting radiotracer used in locating deep brain **tumours** with pinpoint precision.
- **Sodium-24 ($^{24}\mathrm{Na}$)**: Injected as sodium chloride solution to monitor vascular blood circulation velocity and detect thrombosis or occlusions.
- **Gold-198 ($^{198}\mathrm{Au}$)**: Colloidal radioisotope used in intracavitary brachytherapy of neoplastic effusions.
- **Platinum Chemotherapy**: **Cisplatin and Carboplatin** crosslink DNA strands, forming the cornerstone chemotherapy for carcinomas (ovarian, testicular, lung, bladder).
- **Intensity Modulated Radiotherapy (IMRT)**: Computer-controlled linear accelerators manipulate beam intensity to sculpt radiation conformally around irregularly shaped solid tumours, sparing adjacent vital organs.
- **Landmark Heavy Metal Poisonings & Environmental Pathologies**:
- **Minamata Disease (Methylmercury Poisoning)**: First documented in Minamata Bay, Kumamoto Prefecture, Japan (1956). Chemical factory effluent discharged methylmercury into coastal waters, which bioaccumulated through aquatic food chains into fish and shellfish. Human consumption provoked severe neurological devastation: sensory ataxia, peripheral numbness, tunnel vision, dysarthria, and severe spasticity.
- **Itai-Itai Disease ("Ouch-Ouch Disease" / Cadmium Poisoning)**: Mass cadmium poisoning along the Jinzu River basin in Toyama Prefecture, Japan (1912). Mining companies dumped cadmium into river irrigation channels for paddy fields. Cadmium accumulation triggered **severe osteomalacia, multiple pseudofractures, excruciating joint/lumbar pain**, and progressive renal tubular damage.
- **Blue Baby Syndrome (Infantile Methemoglobinemia)**: Consumption of groundwater contaminated with agricultural **Nitrates ($\mathrm{NO_3^-}$)**. Gut microbes in infants reduce nitrates into nitrites, which enter circulation and oxidize the ferrous iron ($\mathrm{Fe^{2+}}$) of hemoglobin into ferric iron ($\mathrm{Fe^{3+}}$), forming **Methemoglobin**. Methemoglobin cannot bind oxygen, producing acute cellular hypoxia and slate-blue skin discoloration (cyanosis).
- **Fluorosis & Knock-Knee Syndrome (Genu valgum)**: Chronic consumption of groundwater with fluoride levels $> 1.5\ \mathrm{mg/L}$ causes dental fluorosis (chalky enamel, yellow-brown mottling) and crippling skeletal fluorosis with **Genu valgum** (inward angling of knee joints causing knees to knock together during walking).

- **Epidemic Dropsy (Argemone Oil Poisoning)**: Ingestion of mustard cooking oil adulterated with the oil of **Argemone mexicana (Mexican prickly poppy)**. Contains toxic isoquinoline alkaloids **Sanguinarine** and dihydrosanguinarine, which uncouple oxidative phosphorylation, cause extensive capillary dilation and hyper-permeability, leading to massive albuminuria, severe pitting peripheral edema (dropsy), glaucoma, and cardiac failure.
- **Silicosis & Pneumoconiosis**:
  - *Silicosis*: Occupational lung fibrosis caused by chronic inhalation of respirable crystalline free silica ($SiO_2$) dust in stone quarrying, quartz grinding, foundry, and tunnel excavation workers.
  - *Pneumoconiosis ("Black Lung")*: Progressive coal-macule fibrosis caused by chronic coal dust inhalation in coal miners.
- **Lathyrism (Neurolathyrism)**: Irreversible spastic paraplegia of the lower limbs caused by habitual, excessive dietary consumption of **Khesari dal (*Lathyrus sativus*)**; mediated by the neurotoxin **$\beta$-ODAP** ($\beta$-N-oxalyl-$\alpha$,$\beta$-diaminopropionic acid).
- **Metabolic, Cardiovascular, Neurological & Sensory Pathology**:
- **Diabetes Mellitus**:
  - *Pathology*: Relative or absolute deficiency of insulin secreted by $\beta$-cells of pancreatic islets of Langerhans, causing chronic hyperglycemia, glycosuria, polyuria, polydipsia, and polyphagia.
  - *Type 1 (IDDM / Juvenile-onset)*: Autoimmune destruction of pancreatic $\beta$-cells.
  - *Type 2 (NIDDM / Adult-onset)*: Peripheral tissue insulin resistance combined with progressive secretory defect.
  - *Management*: Long-term glycemic monitoring via **Glycosylated Hemoglobin (HbA1c)** (reflects preceding 8–12 week mean blood glucose); discovery of insulin by **Banting and Best (1921)**; dietary supplements: **Fenugreek (Methi) seeds** (contain trigonelline and 4-hydroxyisoleucine) and Curry leaves.
- **Diabetes Insipidus ("Water Diabetes")**: Hypothalamic or neurohypophyseal failure to produce or secrete **Antidiuretic Hormone (ADH / Vasopressin)**; distal renal tubules fail to concentrate urine, leading to massive dilute polyuria (up to 15–20 L/day) and relentless unquenchable thirst (polydipsia).
- **Cardiovascular Dynamics & "Golden Hour"**:
  - *Heart Attack (Myocardial Infarction)*: Coronary artery thrombosis following atherosclerotic plaque rupture. **"Golden Hour"** denotes the critical first 60 minutes post-symptom onset during which reperfusion therapy (thrombolysis or primary PCI) rescues maximum ischemic myocardium.
  - *CABG Surgery*: Uses autologous vascular grafts: **Great Saphenous Vein** from leg, **Internal Mammary (Thoracic) Artery** from chest wall, or **Radial Artery** from forearm.
  - *Cath Lab*: Advanced fluoroscopic angiographic operating theater located within the hospital's **Cardiology Department**.
  - *Cardiopulmonary Resuscitation (CPR)*: Emergency life-saving maneuver combining chest compressions (100–120/min) and rescue breathing to maintain coronary and cerebral perfusion during cardiac arrest.

    - *Aspirin*: Acetylsalicylic acid; non-narcotic anti-inflammatory and irreversible inhibitor of platelet cyclooxygenase-1 (COX-1), preventing arterial thrombosis.

- **Neurodegenerative Disorders**:
    - *Alzheimer's Disease*: Progressive cortical dementia marked by cerebral deposition of insoluble **β-amyloid precursor protein plaques** and intracellular hyperphosphorylated **tau neurofibrillary tangles**; first isolated by Alois Alzheimer.
    - *Parkinson's Disease*: Hypokinetic movement disorder caused by selective degeneration of dopaminergic neurons in the substantia nigra; characterized by resting tremor, rigidity, bradykinesia, and postural instability. **Arvid Carlsson** won the 2000 Nobel Prize for discovering dopamine's neurotransmitter function and establishing L-dopa replacement therapy.

- **Ophthalmic Refractive Errors & Topography**:
    - *Myopia (Near-Sightedness)*: Eyeball anteroposterior diameter elongated; image focuses in front of retina; distant objects blurred; corrected using **Concave (diverging) spherical lenses**.
    - *Hypermetropia (Far-Sightedness)*: Eyeball shortened; image focuses behind retina; near objects blurred; corrected using **Convex (converging) spherical lenses**.
    - *Astigmatism*: Asymmetric meridian curvature of cornea or crystalline lens; light refracted unevenly at different planes; corrected using **Cylindrical lenses**.
    - *Cataract*: Senile denaturation and aggregation of lens crystallin proteins creating a milky cloudy opacity; surgically managed by phacoemulsification and **Intraocular Lens (IOL)** implantation.
    - *Retinal Landmarks*: **Fovea Centralis (Yellow Spot / Macula lutea)** possesses exclusive packing of cone photoreceptors, providing highest visual resolving power (sharpest acuity); **Optic Disc (Blind Spot)** is completely devoid of photoreceptors at the exit site of the optic nerve.
    - *Mydriatic Alkaloid*: Ophthalmic drops of **Atropine** (extracted from *Atropa belladonna*) block pupillary constrictor muscarinic receptors, causing sustained pupil dilation.

- **Other Clinical Disorders**:
    - *Oral Submucous Fibrosis (OSF)*: Pre-malignant chronic fibrotic scarring and blanching of the oral submucosa; caused by regular chewing of **Areca nut / Gutkha** containing arecoline alkaloid; leads to progressive lockjaw (inability to open mouth).
    - *Lukoskin*: Dual polyherbal formulation (liquid + ointment) developed by DRDO's **Defence Institute of Bio-Energy Research (DIBER), Haldwani** for the therapeutic management of Vitiligo (Leucoderma).
    - *Cerebral Palsy*: Non-progressive congenital movement and postural neuro-disorder caused by hypoxic/ischemic brain damage during perinatal encephalic development in young children.
    - *Bulimia Nervosa*: Eating disorder characterized by episodes of uncontrolled binge eating followed by self-induced vomiting or laxative purging.
    - *Sleep Apnea*: Repetitive cessation of respiration during sleep caused by upper pharyngeal airway collapse; hallmark symptom is heavy loud snoring.
    - *Anosmia*: Total or partial loss of the olfactory sense (smell); common in chronic rhinitis, post-viral trauma, and cribriform plate head injuries.

## 3.7 Vaccines, Diagnostic Tests & Clinical Instruments

- **Immunology Foundations: Active vs Passive Immunity**:
- **Active Immunity**: Host's own immune system is stimulated to produce antibodies and memory lymphocytes following natural infection or vaccination; slow onset, long-lasting / lifelong protection.
- **Passive Immunity**: Ready-made pre-formed antibodies are directly introduced into the body; produces instantaneous protection but confers short-lived immunity with no immunological memory:
  - Natural passive: Maternal **IgG** crossing the placenta to the fetus; secretory **IgA** delivered to newborns via maternal **Colostrum** (initial yellowish breast milk).
  - Artificial passive: Anti-tetanus serum (ATS), Anti-rabies serum, Anti-snake venom (ASV).
- **The Global COVID-19 Vaccine Arsenal**:

| Vaccine Trade Name | Developing Manufacturer / Country | Vaccine Technology Platform | Antigen Used |
|---|---|---|---|
| **Covaxin (BBV152)** | Bharat Biotech & ICMR (India) | **Inactivated Whole Virus** | Chemically killed whole SARS-CoV-2 virion |
| **Sinovac (CoronaVac)** | Sinovac Biotech (China) | **Inactivated Whole Virus** | Chemically killed whole SARS-CoV-2 virion |
| **Covishield** | Serum Institute of India / AstraZeneca | **Non-Replicating Viral Vector** | Modified Chimpanzee Adenovirus (ChAdOx1) carrying S-gene |
| **Sputnik V** | Gamaleya Research Institute (Russia) | **Heterologous Viral Vector** | Recombinant Human Adenoviruses (Ad26 & Ad5) |
| **Moderna (mRNA-1273)** | Moderna (USA) | **Synthetic mRNA in LNPs** | mRNA encoding stabilized Spike glycoprotein |
| **Pfizer-BioNTech** | Pfizer & BioNTech (USA/Germany) | **Synthetic mRNA in LNPs** | mRNA encoding full-length Spike glycoprotein |
| **Corbevax** | Biological E. Limited (India) | **Protein Subunit** | Recombinant Spike protein receptor-binding domain |
| **ZyCoV-D** | Zydus Cadila (India) | **Plasmid DNA (Needle-free)** | Plasmid DNA expressing spike protein; Tropis applicator |

- **Comprehensive Medical Diagnostic Tests & Instruments**:
- **Widal Test**: Tube/slide agglutination test for **Typhoid Fever** (*Salmonella typhi*).
- **Mantoux Test**: Intradermal injection of 5 TU Purified Protein Derivative (PPD); read at 48–72 hours for **Tuberculosis**.
- **Schick Test**: Intradermal toxin test assessing susceptibility to **Diphtheria**.
- **VDRL / RPR / Wassermann Test**: Screening serological tests for **Syphilis** (*Treponema pallidum*).
- **ELISA (Enzyme-Linked Immunosorbent Assay)**: Rapid enzyme-substrate screening assay for **HIV/AIDS**, Hepatitis B, and hormones.
- **Western Blotting**: Confirmatory antibody-protein detection test for **HIV/AIDS**.

- **RT-PCR (Reverse Transcription Polymerase Chain Reaction)**: Molecular detection of viral RNA in **COVID-19, Swine Flu (H1N1), and Hepatitis C**.
- **ECG (Electrocardiogram)**: Records cardiac electrical conduction: **P Wave** (atrial depolarization), **QRS Complex** (ventricular depolarization), **T Wave** (ventricular repolarization). (Willem Einthoven).
- **EEG (Electroencephalogram)**: Records brain wave frequencies: Alpha (awake relaxed), Beta (active alert), Theta, Delta (deep sleep). (Hans Berger).
- **Endoscope**: Flexible fiber-optic tube used for direct visual examination of hollow internal viscera (stomach, duodenum, colon); functions entirely on the physical principle of **Total Internal Reflection (TIR)**.

## One-Liner Revision -- Nutrition, Diseases and Medicine

| # | Fast Fact |
|---|---|
| 1 | Carbohydrate = primary energy source; **4 kcal/g** |
| 2 | Fat = most concentrated energy source; **9 kcal/g** |
| 3 | Protein = **4 kcal/g**; essential amino acids = 9 (must come from diet) |
| 4 | Vitamin A (Retinol) = Night blindness; sources: carrot, liver, egg |
| 5 | Vitamin B1 (Thiamine) = Beriberi; found in rice bran, cereals |
| 6 | Vitamin B2 (Riboflavin) = Ariboflavinosis (cracks at mouth corners) |
| 7 | Vitamin B3 (Niacin) = Pellagra (4Ds: Dermatitis, Diarrhoea, Dementia, Death) |
| 8 | Vitamin B9 (Folic acid) = Neural tube defects in foetus if deficient |
| 9 | Vitamin B12 (Cobalamin) = Pernicious anaemia; only in animal foods |
| 10 | Vitamin C (Ascorbic acid) = Scurvy; richest source = Amla (Indian gooseberry) |
| 11 | Vitamin D (Calciferol) = Rickets (children), Osteomalacia (adults); sunlight synthesis |
| 12 | Vitamin E (Tocopherol) = antioxidant; deficiency = sterility/muscle weakness |
| 13 | Vitamin K (Phylloquinone) = blood clotting; deficiency = excessive bleeding |
| 14 | Fat-soluble vitamins = **A, D, E, K** (stored in liver/fat) |
| 15 | Water-soluble vitamins = **B-complex + C** (excreted in urine; daily needed) |
| 16 | Iron deficiency = **Anaemia** (haemoglobin low; pale, fatigued) |
| 17 | Iodine deficiency = **Goitre** (enlarged thyroid) |
| 18 | Calcium deficiency = **Tetany / Osteoporosis** |
| 19 | Fluoride excess = **Fluorosis** (mottled teeth, skeletal damage) |
| 20 | Penicillin discovered = **Alexander Fleming (1928)**; Nobel 1945 (Fleming+Chain+Florey) |

| # | Fast Fact |
|---|---|
| 21 | First antibiotic = **Penicillin** (from *Penicillium notatum* mould) |
| 22 | Malaria vector = **Female Anopheles mosquito**; caused by Plasmodium |
| 23 | Dengue vector = **Aedes aegypti** mosquito |
| 24 | Filaria (Lymphatic filariasis) vector = **Culex mosquito** |
| 25 | Plague (Black Death) vector = rat flea (*Xenopsylla cheopis*); bacterium *Yersinia pestis* |
| 26 | Cholera = *Vibrio cholerae*; rice-water stools; foul-smelling; water-borne |
| 27 | Tuberculosis = *Mycobacterium tuberculosis*; Mantoux test; BCG vaccine |
| 28 | Vaccine for smallpox = **Edward Jenner (1796)**; smallpox eradicated **1980** (WHO) |
| 29 | ORS (Oral Rehydration Solution) = key treatment for diarrhoea; WHO/UNICEF |
| 30 | DOTS = Directly Observed Treatment Short-course; for **tuberculosis** control |